Use-Case Flow Specification

Dhruv Swatantramath | PES1UG24CS155 | Warehouse Inventory & Pallet Location Tracker | Problem Statement #27

Use Case	Assign Pallet to Shelf Location
Primary Actor	Warehouse Operator
Related FR(s)	FR-001, FR-003
Preconditions	The pallet has already been scanned into the system (barcode/RFID read worked). The Warehouse Operator is logged in. The system already knows the rack layout and how much load each rack is currently carrying.
Postconditions	If it works: the pallet's location is updated, the rack's load total includes the new pallet, and a movement log entry is created. If it fails: the pallet's location stays the same and no log entry is added.

Main Success Scenario

1. The Warehouse Operator picks "Assign Pallet to Shelf Location" once a pallet has been scanned in.
2. The system asks the operator to enter or confirm which shelf the pallet should go to (aisle, rack, bin).
3. The operator submits these coordinates.
4. The system checks the pallet's weight against the rack's limit, using the Validate Weight Capacity use case (this includes any pallets already on that rack).
5. The system confirms the pallet can be placed there without going over the weight limit.
6. The system saves the pallet's new location along with a timestamp.
7. The system logs this as a stock movement, noting the operator, the pallet ID, and where it was placed.
8. The system shows a confirmation message with the pallet ID and its new shelf location.
9. Use case ends successfully.

Alternate Flow

4a. Weight limit exceeded

4a1. The system finds that this placement would push the rack over its weight limit.

4a2. The system rejects the placement and tells the operator by how much the limit would be exceeded.

4a3. If a nearby rack has room, the system suggests it as an alternative.

4a4. The operator either picks a different rack (going back to step 2) or cancels — either way, nothing is saved.